

The combination of forecasts in the trading of electricity from renewable energy sources

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Published online: 27 April 2013
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Abstract The legal support provided by the ‘Act on Granting Priority to Renewable Energy Sources’ (German *Erneuerbare Energien Gesetz*, EEG) and its precursor has in the last 20 years led to a marked growth in Germany in the use of renewable energies to generate electricity. As a result of the EEG amendment adopted in the summer of 2011 and in force since 1 January 2012, the market integration of electricity generated from renewable energy sources (RES-E) has become more important. Consequently, the economic importance of trading RES-E has also increased. A major role in determining costs in trading electricity from wind and solar energy on the wholesale markets plays the forecasting method used. If a forecast inaccurately predicts the amount of electricity actually generated, one result could be elevated costs in the trading process. In the beginning of this article we introduce the legal framework governing the trading of RES-E. Subsequently, we present a method for combining several individual forecasting methods. Finally, using empirical data, we show that in comparison to the best available individual forecast, the proposed combined forecast results in a clear improvement of forecasting quality as well as in a reduction in trading costs.

Keywords Renewable energy sources · Combination of forecasts · Electricity markets · Power exchange trading

JEL Classification Q47 · C53 · C22

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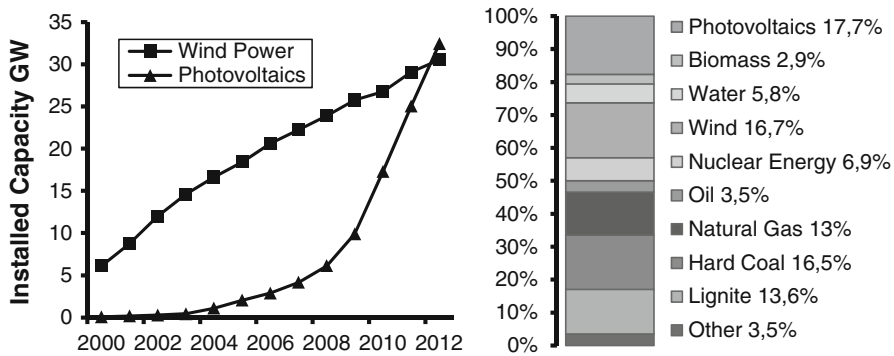


Fig. 1 Growth of renewable energies/power plant capacity by primary energy source 2012

1 Introduction

In the last 20 years, the generation of electricity from renewable energy sources (RES-E) has greatly increased in Germany. The main reason for this development is the social and political will to increase the proportion of RES-E and the resulting legal support scheme. The legal framework for today's support is the in the year 2000 established Renewable Energy Sources Act (EEG) which was recently amended in 2011 (on the origin of the EEG see, for example, Bruns et al. 2010, pp. 61ff). Most importantly, the EEG provides financial support and, in comparison to conventional power plants, the priority feed-in into the electricity grid of RES-E.¹ Especially on the part of private investors, a very high willingness to invest was a result of these legal regulations; wind energy and photovoltaic plants experienced an especially robust growth. Altogether, the installed power plant capacity of renewable energy by the end of 2012 amounted to—at almost 79 gigawatts (GW)—43 % of the total installed capacity in Germany. Figure 1 shows the growth of renewable energy in the last 10 years and the percentage of the primary energy sources of the German power station capacity. All capacities correspond to the standing at the end of 2012 (see IWES 2013 for wind, and Bundesnetzagentur ('Federal Network Agency') 2013 for PV; all other energy sources are estimated based on Bundesministerium für Wirtschaft und Technologie 2012).

Because of the high percentage of installed power station capacity, renewables already have a large influence on the energy supply system. Thus in 2012, 21.9 % of electric power consumption was covered by renewables (see AG Energiebilanzen 2012). In particular the electricity grids, originally not designed for the distributed generation of electricity, are thus affected (see Winter et al. 2011). Moreover, the heavily fluctuating generation from renewables has also a significant influence on the electricity market.

¹ In 1991 important features of the current state support were already introduced by means of the Act on the Feed-in of Electricity from Renewable Energy Sources into the Grid; short title, Feed-In Act (see Jarass et al. 2009, p. 98).

With the version of the EEG that was valid up to the end of 2011, a direct marketing of RES-E was only in a few cases economically practical. That is why to a great extent the responsible network providers integrated this source of electricity into the electricity markets. The EEG amendment enacted in the summer of 2011 and in effect since 1 January 2012 introduced an option as of 2012 for direct marketing that for a large number of plant operators has proven attractive. The second section of this essay will provide an overview of the various options inherent in the market integration of RES-E afforded by the EEG, and in a subsequent part of the same section the course of this integration and the resulting costs will be sketched.

In any discussion of the trading of fluctuating RES-E, such as wind and sun, special attention must be paid to the fact that its supply depends upon factors that for the operator of a plant can scarcely be controlled. For the purposes of forecasting this supply, various procedures with different advantages and disadvantages have been developed (see Giebel et al. 2011). Naturally, the costs of trading RES-E largely depend upon the quality of the forecast; the better the forecast, the fewer the expensive, compensatory transactions need to be taken on the electricity markets. In the third section of this study it will be shown, by means of a combination of forecasts, how as compared to even the best individual forecast, an improvement of forecast quality and a cost reduction is possible. In order to achieve this end a method of combining several forecasts, whose ultimate goal is to minimise the costs of trading RES-E, will be developed and presented on the basis of empirical data.

2 Trading of electricity from renewable energy sources

The trading of electricity from fluctuating renewable energies differs in two important elements from the trading of electricity from conventional power plants. The first difference is the state support of RES-E in accordance with the EEG. In the integration of the EEG-subsidised electricity in the normal competitive electricity markets, extensive legal regulations have been developed and must be taken into account in the trading of RES-E. The second difference is the so-called supply-dependent production. In contrast to conventional power plants that, for example, work with fossil fuels, the supply of renewable energies cannot be influenced by the producers. Thus in the trading of the renewables a forecast of the supply of energy is necessary. Forecasting errors have to be compensated by means of an expensive procedure, which can lead to high costs. Following a short overview of the possible forms of market integration of RES-E in Sect. 2.1, this procedure will be presented in Sect. 2.2 and the interrelation between forecasting errors and costs in Sect. 2.3.

2.1 Market integration of RES-E

The RES-E supported by the EEG can as of 2012 be integrated into the competitive electricity market by means of various alternatives. The first option, up to the present almost the only one exercised, is market integration via the network operator. The plant operator in turn receives the legally guaranteed feed-in tariff through the network operator, and the trading then proceeds without the operator's

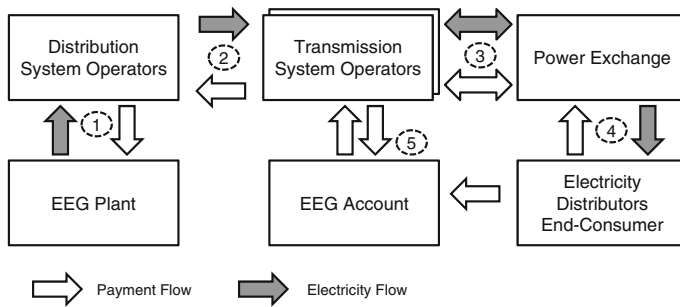


Fig. 2 Market integration of renewable energies via the grid operators

participation in any form. Additional options are aligned to the direct marketing of renewable energy; here the plant operators or their service providers trade the electricity produced.

The market integration of renewable energy via the grid operator follows an elaborate process and is regulated by the EEG and the Ordinance on the Further Development of the Nationwide Equalisation Scheme (*AusglMechV*; 'Equalisation Scheme Ordinance'). An overview of the process is presented in Fig. 2.

In the first step of the process the electricity generated in an EEG plant is fed into the grid of the responsible network operator, who usually is a distribution system operator (see §5ff EEG). The EEG plant operator receives from the network operator the guaranteed feed in tariff appropriate to the plant (see § 16ff EEG). In the second step the sale of the total electricity generated according to the EEG in the network of a distribution system operator is delivered to the appropriate transmission system operator (TSO).² The sale price corresponds to the average of the feed in tariffs paid to the plant operators. In the third step the TSO sells the electricity on the power exchange (see § 2 'Equalisation Scheme Ordinance' in combination with § 64 EEG). The TSO receives the regular market price for the electricity sold. In the fourth step the electricity is passed on to the end-users via the various electricity distributors on the exchange. Since the market price on the average is much lower than that of the feed in tariff for EEG electricity, as a rule a deficit results for the TSOs. As described in the fifth step, a special EEG levy is imposed on the end consumer in order to compensate for this deficit (see § 3 'Equalisation Scheme Ordinance').³

For the direct marketing of RES-E, there are according to the version of the EEG that came into force in 2012 two completely different options (see § 33b EEG). The first is the direct marketing for the reduction in the EEG levy. The basic idea here is that an electricity supply company is permitted to reduce the EEG levy if specific proportions of the electricity delivered to the customer were derived from specific

² Germany is currently divided into four regional zones, each with a responsible TSO. The four TSOs exchange the EEG electricity among themselves in order to balance the uneven distribution of the EEG plants in the German Federal Republic (see § 34ff., EEG).

³ In 2011 the levy amounted to 3.53 cents for each KWh consumed, in 2012 it was 3.592 cents for each KWh and in 2013 it is 5.277 cents for each KWh. Basically, every electricity end-user must pay the levy. Exceptions are especially energy-intensive companies.

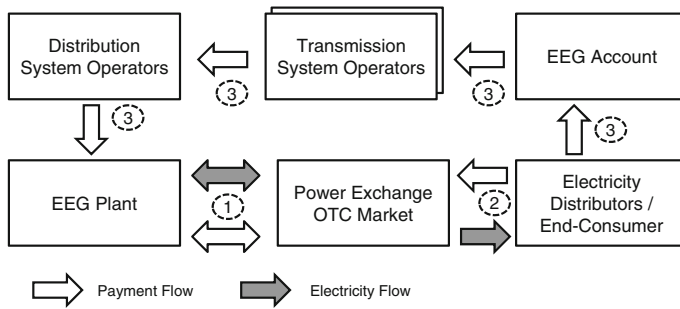


Fig. 3 Market integration of renewable energy through direct marketing

plants subsidised by the EEG (see § 39 EEG). Since this option for direct marketing is only of interest under very limited conditions, it will not be further discussed in this article.

The second option is based on direct marketing under the claim of a market premium (see § 33 g EEG).⁴ Figure 3 depicts an overview of the process of this form of marketing.

In the first step a plant operator or a service provider on behalf of the plant operator sells the generated electricity on the electricity exchange or on the over-the-counter (OTC) market. For this transaction the plant operator receives the regular market price. In the second step the electricity is passed on to the end-users via the various electricity distributors. In the third step the plant operator obtains from the network operator in addition to the market revenues a market premium from the EEG account. This bonus is calculated *ex post* once a month and is comprised of two parts. The first part is the difference between a reference market value and the alternative feed in tariff of a concerned plant. The reference market value is for all plants of a given energy source the same and is calculated on the basis of the average of their quarter-hourly production and the price on the power exchange. The second part, the management premium, is intended to cover the trading costs of the plant operator and is paid as a regular fee that differs depending on the energy source. If the trading occurs exclusively via the power exchange, that is, at the same price as in the calculation of the reference market value, only the risk remains for the plant operator that the marketing costs exceed the management bonus.⁵ If on the basis of an especially cost-effective trading strategy, the trading costs drop significantly below the value of the management premium, this form of direct marketing can be of significant economic interest.

⁴ Some years ago a market premium was introduced in other European countries, for example, in Spain and Denmark. The idea for a moving market premium for the direct merchandising by the plants supported by the EEG in Germany goes back to the Fraunhofer Institut für System- und Innovationsforschung in Karlsruhe ('Fraunhofer Institute for Systems and Innovations Research') (see, for example, Sensfuss, Ragwitz 2009).

⁵ Since the market premium is calculated on the basis of the average production of all plants of a specific source of renewable energy, a risk remains when the individual plant deviates in its delivery profile from the average value. As a result the sum from the moving market premium and power exchange revenues can turn out to be slightly higher or lower than the EEG compensation.

2.2 Trading procedure

The following remarks will first explain the process of trading RES-E as it corresponds to the third step in Fig. 2, using the example of a TSO, since at the present time large portions of renewable energy are by this means integrated into the electricity market. Subsequently, on this basis the differences in direct marketing will be briefly sketched.

For the TSO there is a series of regulatory guidelines that must be observed in trading RES-E. They are laid out in the Ordinance on the Further Development of the Nationwide Equalisation Scheme ('Equalisation Scheme Ordinance'). In addition, there are economic incentives for cost-effective marketing.⁶

The trading of RES-E via the TSO must occur exclusively on the day-ahead spot market of an power exchange (compare § 1, paragraph 1, 'Equalisation Scheme Ordinance'). At the present time the only available day-ahead spot market in Germany is the European Power Exchange (EPEX).⁷ The day-ahead market of the EPEX is organised as a uniform price auction; that is, at auction a uniform marginal price is established (for a detailed presentation of various auction forms, see Grimm et al. 2008; for details of the power exchange dealings, compare Schiffer 2010, p. 292). Here as the smallest product size twenty-four individual hour blocks are tradable; that is, in each case the constant delivery or acceptance for 1 h of a specific amount of electricity. The gate closure for the last possible bids is 12:00 day-ahead. In accordance with the 'Equalisation Scheme Ordinance', this means that the TSOs must at the very latest just prior to gate closure submit a bid based on forecasts for the total EEG electricity to be marketed regardless of price. In order to avoid extremely negative prices, in exceptional cases a limitation on the bids is permissible (see § 8 'Equalisation Scheme Ordinance' 2010; for a discussion of negative prices, see, among others, Andor et al. 2010). In Fig. 4 the day-ahead trading of wind-power for 5 March 2011 is shown.

For the day-ahead marketing the forecasting horizon at the gate closure at 12:00 is a maximum of 36 h in addition to the time required for the bid. If the forecasting horizon is reduced, the quality of the forecast increases. Therefore, differences occur between a day-ahead forecast and an intraday forecast, which has a shorter forecasting horizon, or differences between intraday forecasts with a different

⁶ The TSOs are instructed by legislators to carry out the trading as efficiently (see § 64 Paragraph 3, EEG), transparently and objectively (see § 2 'Equalisation Scheme Ordinance') as possible and in accordance with the latest state of science and technology (see §§ 1, 3 'Equalisation Scheme Ordinance'). In addition, for the TSO an incentive mechanism for EEG marketing was developed. The mechanism for an efficient trading is regulated in § 7 der 'Equalisation Scheme Ordinance' and is based upon a consideration of costs. If the costs of the EEG trading decrease per marketed unit in comparison to a reference value, the TSO receives a bonus in the amount of 25 % of the costs saved.

⁷ In accordance with the 'Equalisation Scheme Ordinance' it is also possible for a TSO to trade on a stock exchange in another market area in Europe, but the transport of the electricity abroad and the reservation of the necessary transport capacity by means, for example, of explicit capacity auctions, are more complex and carry additional risks. Moreover, the integration of the European domestic market for electricity through the current development of market coupling is increasing, so that a decline of price differences between the European market areas is to be expected (see A. Weber et al. 2010, pp. 303ff.). Thus in practice all German TSOs exclusively employ the EPEX for their marketing activities.

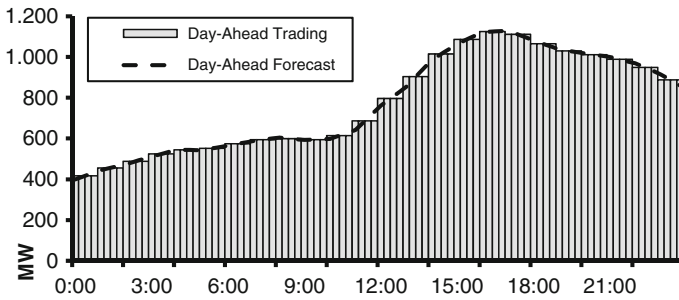


Fig. 4 Day-ahead trading and forecasting of wind-power for 5 March 2011 (data source: TransnetBW GmbH)

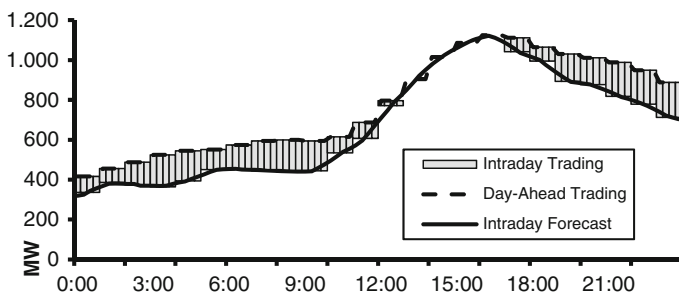


Fig. 5 Intraday forecasting and intraday trading of wind energy on 5 March 2011 (data source: TransnetBW GmbH)

forecasting horizon. The balancing of these forecasting differences is in accordance with the ‘Equalisation Scheme Ordinance’ possible via an intraday market of an power exchange (see § 1, Paragraph 2, ‘Equalisation Scheme Ordinance’). Like the day-ahead marketing, here the platform of the EPEX is used by the TSOs. The intraday spot market is in contrast to the day-ahead spot market organised by continuous trading. The trading of hourly products is here possible around the clock as of 15:00 the day before until forty-five minutes before delivery (see Barquín et al. 2011, pp. 12ff). Since the year 2012 the trading of quarterly-hour products is also possible on the intraday market. The auction mechanism in the intraday market is in analogy with many other exchange markets organised by pay-as-bid pricing. For a specific hourly product there can be, consequently, and depending on the time of trading, differing prices. Figure 5 shows the intraday trading of wind energy.

Forecasting errors that remain after the closing of intraday trading can no longer actively be balanced according to the ‘Equalisation Scheme Ordinance’. For this balancing the mechanism of balancing energy must be used.⁸ In the process a price

⁸ Control reserves are required in the electricity network to continuously maintain a balance between supply and demand of electricity. By means of control reserves, power supply companies or electricity producers are offered balancing energy to offset short-term demand or production fluctuations. For this purpose power stations, which on short notice can increase or decrease production, receive contracts from TSOs according to a regulated process. The costs for the compensation of the power stations are paid by the TSOs.

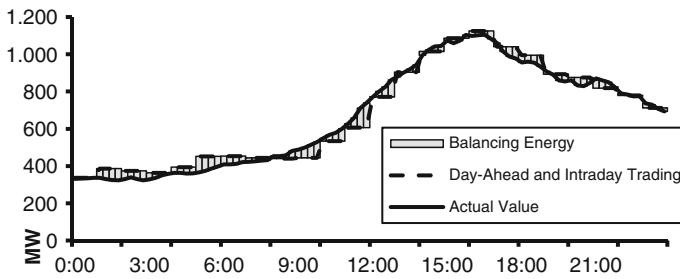


Fig. 6 Actual-value of the wind energy and balancing energy on 5 March 2011 (data source: TransnetBW GmbH)

becomes due, the so-called balancing energy price, that depends upon the given market situation (see Swider 2006, pp. 5ff).⁹ Figure 6 depicts the balancing energy used for 5 March 2011 in the trading of wind energy.

In the direct marketing according to the market premium scheme, the legally valid guidelines governing the TSOs are not binding. Because of the calculation of the reference market value on the basis of the spot market price, a day-ahead marketing on the electricity exchange or at least on the basis of the exchange reference price is, however, appropriate in order to avoid price risks (see Sect. 2.1). In contrast to day-ahead trading, intraday exchange trading assumes no special role. Thus for the balancing process in the intraday market, in addition to the exchange, the OTC markets can also be used without additional price risks. The balancing energy mechanism in turn is identical for all market players. Differences between direct marketing and marketing by means of the TSO do not, therefore, exist there. Likewise there are in both forms of the marketing no significant differences in regard to the purpose of trading RES-E with the smallest costs: in the case of direct marketing the trading costs have a bearing directly on the company profits, in the case of marketing through the TSO indirectly via the incentive mechanism. In the next section an outline of the trading costs and their correlation with the quality of the forecast will be presented.

2.3 Costs of RES-E trading

The costs of RES-E trading are principally caused by the intraday and real-time balancing of forecasting differences. The prices on the intraday market and for balancing energy are very volatile and are strongly influenced by the trading of RES-E which leads to high transaction costs (see Weber 2010, p. 3156). The volatility of the prices is caused by the low liquidity on the intraday market, especially at night and on weekends. In comparison to the volume on the day-ahead market, the liquidity on the intraday market in 2009, for example, amounted to only

⁹ The balancing energy price is always the same for all market participants. In the event of an excess demand for electricity, the price is typically more expensive than on the spot market; in the case of an excess supply, electricity is typically less expensive. Thus for market players costs occur when their needs for balancing energy are adjusted to the needs in total.

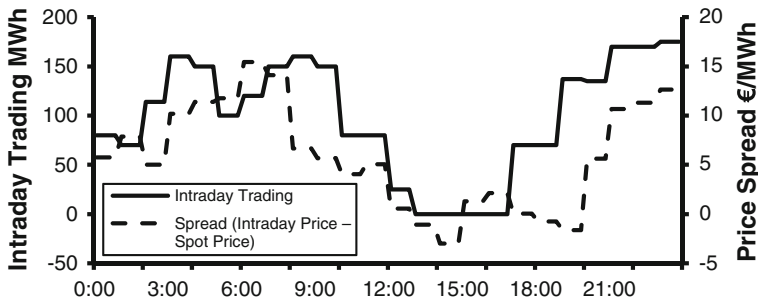


Fig. 7 Influence of the trading activities on the price on the intraday market on 5 March 2011 (data source: TransnetBW GmbH)

just under 4 % (see Schiffer 2010, p. 293). Figure 7 shows on the example of 5 March 2011 the influence of the renewable energy on the price on the intraday market. The example is typical of a situation characterised by middle to high forecasting differences.

In Fig. 7 the trade volumes of the TransnetBW GmbH on the intraday market of the EPEX are given by the solid line. The price spread between the average price of the intraday trading and the unit price of the day-ahead trading is illustrated on the secondary axis, the dotted line. In the graph a strongly positive correlation between the two values can be recognised. In Fig. 8 this correlation is represented in the graph on the left by means of an analysis of a data set from the TransnetBW GmbH that is based on all the intraday transactions for 1 year for balancing the error of wind power forecasts. The data record contains for every hour of the time period in which the intraday transactions were executed the respective total purchases and sales as well as the corresponding mean price. It is noticeable that there is a positive relation between the amount of the transactions (i.e., the forecast difference) and the price spread. This relation can be explained with a coefficient of determination of 0.335 through a linear regression with an increase of 0.049.

The product from the transaction volume and the appropriate spread between the transaction price and the day-ahead price can be considered as the cost of the

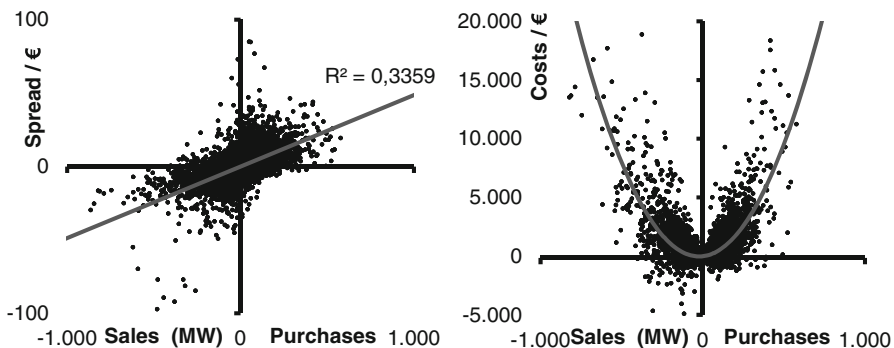


Fig. 8 Correlation between difference in forecast and price spread or difference in forecast and costs

forecast difference. Here the day-ahead price is considered as independent. In the right-hand graph in Fig. 8 the correlation between the differences in forecasting and these costs is illustrated. On the basis of the positive correlations between forecast difference and price spread, the correlation between difference in forecast and costs can be approximated by means of a quadratic function.

Comparing the average prices of purchases and sales transactions of the TransnetBW GmbH for the balancing of the differences in wind forecasting, this correlation can likewise be seen. In August 2010 the spread in the two prices amounted, for example, to 9.90 €/MWh. By means of this spread the cost for the balancing of the forecast differences can be calculated. Doing that it is assumed that the market prices at the forecasting point are independent of the forecasting differences known only at a later point in time. In the case of unbiased forecasts for the spot and intraday trading, purchase and sale volumes, correspond to each other on the intraday market. Through the purchase and sale of one MWh, a transaction volume of two MWh occurs, and on the average costs in the amount of the simple price spread arises. The costs of the total intraday balancing of forecast differences thus correspond under this assumption exactly to half of the total transaction volume multiplied by the price spread. Even if an ideal unbiasedness in the forecasts in a finite period of time does not entirely exist, it can nevertheless be shown empirically that with the aid of the price spread a very good estimate of the costs is possible.

Just as the prices on the intraday market are affected through the differences in forecasting, the balancing energy prices are influenced by the forecasting errors remaining after the intraday trading. On the basis of the small number of power plants used for providing control reserves, the merit-order curve in the case of balancing energy, however, is steeper than that in the intraday market. Thus transactions of the same volume cause on the average larger price changes in the case of balancing energy than on the intraday market. The correlation between the price spread of balancing energy- and spot-market price and the transaction volume can again be demonstrated by means of a linear regression. With a slope of 0.162 the regression line is more than three times steeper than the regression line represented in Fig. 8 for the intraday market. Therefore, the difference between the average purchase and sale price is clearly higher. In the case of the TransnetBW GmbH, this difference amounted in August 2010 in the trading of wind energy, for example, to 36.96 €/MWh.

Thus the costs of RES-E trading are influenced by the spread between the purchases and sales prices on the intraday market and by spread in the case of balancing energy as well as in the respective volume of the transactions. The decisive factor of influence on the costs of the RES-E trading is thus the quality of the forecasts used for trading activities. The quality of the forecasts and thus the amount of the forecasting differences have a direct influence on the transaction volume and via the price effect also an influence on the price spread. An improvement in the quality of the forecasting thus leads directly and indirectly to a reduction in costs. Besides the quality of the forecast the volatility of the forecasts in dependence on the forecast horizon is of great importance. Since forecasting differences on the intraday market are continuously balanced, fluctuations in the forecasting can with a decreasing forecast horizon lead to counter transactions and

thus to a higher trading volume. This increased trading volume can cause high transaction costs. On this point a detailed illustration will be presented in Sect. 3.5. An additional important factor of influence on the trading costs is the time a transactions is carried out on the intraday market. The shorter the time period between the day-ahead trading and the intraday trading, the smaller is the risk of a high price spread.

3 A method for the combination of renewable energy forecasts

In the last two decades different approaches for forecasting renewable energy, such as wind and solar energy, have been developed. The basis for most of the approaches are numerical weather models that on the basis of the given conditions of the atmosphere compute the weather conditions for the next few days by means of mathematical models (see Giebel et al. 2011, p. 5). As a result forecasts on various weather parameters, such as wind speed, cloud cover and precipitation, are available in a more or less rough grid. Numerical weather models are run mostly by state weather services, for example the German Weather Service. Often the model is run by the Weather Service several times a day (see Lange and Focken 2005, pp. 8ff).

For the power forecasts for wind power or photovoltaics, i.e., the conversion of the meteorological parameters in terms of power output figures, statistical and physical approaches can be differentiated (see Giebel et al. 2011, p. 5). Fundamental correlations between meteorological variables and power generation are the basis of the physical approaches. For that purpose, the results of the numerical weather models must first be refined in order to calculate, for example, the wind speed at the respective plant location. Then, with the aid of the power curves of the wind power plants, the electrical output can be calculated (see Jarass et al. 2009, pp. 28ff). Physical approaches can achieve very good results, but require an exact refining of the local meteorological parameters and extensive data on the power plants and plant locations. In the statistical approach, for example, methods such as artificial neural networks or multiple regression models are used (see Lange and Focken 2005, pp. 10ff). The advantage of this approach is that a modelling with less detailed data on power plant and plant location is possible. Thus statistical approaches are easier to implement, but require extensive historical data. On the basis of the here briefly summarised differing advantages of the respective approaches, combinations of various approaches are, in practice, often used.

Several commercial providers offer power forecasts in Germany. Through a variety of companies a strong competition has arisen. Therefore, in the following section suitable commercially available power forecasts are relied upon in order to develop a forecasting method that leads to the smallest trading costs. The goal of this forecasting method is to choose the power forecasts and to combine them in such a way that for each point in time the best available forecast or combination of forecasts is used. Since, in addition, the available generation capacities in real time (the actual-value) can improve forecasting quality, these are employed additionally for a short-term forecast horizon. Figure 9 schematically depicts the development of the forecasting method. It combines available forecasts and takes into consideration, where applicable, additional actual-values. The arrows indicate a possible data flow.

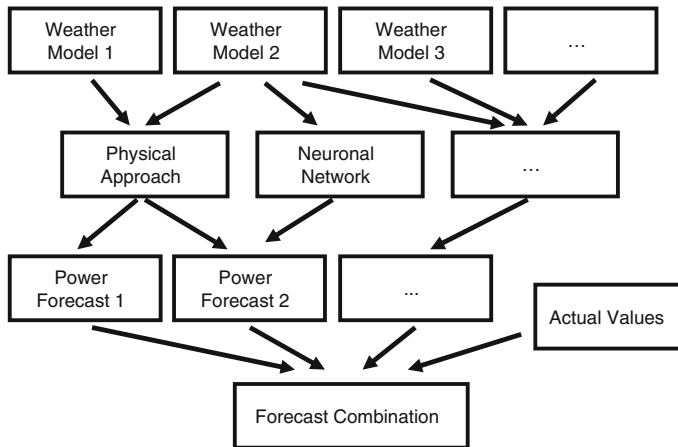


Fig. 9 Schematic representation of the forecasting method

3.1 The combination of forecasts

The combination of various quantitative forecasts for improving the forecasting quality has for a long time been used in many research disciplines (see Clemen 1989, p. 560). The first systematic discussion with this purpose is attributed to a study in which the combination of various methods for the forecasting of airline passenger miles flown led to a smaller variance in the forecasting error compared to the best individual forecast (see Bates and Granger 1969). Since then a vast number of different approaches has been suggested and analysed to combine forecasts (for a survey see, for example, Timmermann 2006). For many empirical issues, such as forecasts of inflation rates (see, e.g., Stock and Watson 2003 or Wright 2009), stock prices (see, e.g., Clemen 1989, p. 565), or weather data (see, e.g., Krishnamurti et al. 2000 or Stephenson et al. 2005), appropriate approaches were successfully employed.

The basic idea in combining forecasts is that each of the individual forecasts employed contains unique and useful information to achieve the goal of the forecast (see Thiele 1992, p. 12). This unique information can, for example, be based on different methods or sources of information. For a decision-maker having a stake in a specific variable of interest for which several individual forecasts are available, the basic question arises whether a combination of forecasts or an own forecasting method that is based on all the individual bits of information should be employed. Looked at from the theoretical standpoint, the second approach appears better, since bits of information get lost in the course of individual forecasts. In the case of complex models, especially of one with a small sample in relation to the number of parameters to be estimated, a combination, however, can be advantageous (see Timmermann 2006, pp. 143ff). In practice, moreover, additional reasons often recommend a combination of forecasts. Firstly, often some individual information is not available, or only at high costs, or, secondly, the time needed for the

accumulation of all the individual bits of information is in the course of daily business operations too large (see Diebold and Pauly 1990, p. 503). Thirdly, the risk of outage declines in business operations through the use of several forecasts. All three arguments apply to the matter under discussion.

A majority of the methods developed for the combination of forecasts is based on a combination for minimising the mean squared forecast error (see Timmermann 2006, p. 142). Often a method based upon regression analysis is used (for a survey see Thiele 1992, p. 53). The original approach by Bates and Granger, called the variance–covariance method, can be also be assigned to this approach. In empirical studies it could be demonstrated that simpler methods, such as equal weighting or a combination based exclusively on the variances of the individual forecasts, can also yield good results. Thereby it has also been shown that methods by which only a few parameters must be estimated are often superior to more elaborate ones (see Timmermann 2006, pp. 181ff). A combination of wind power forecasts for the improvement of the quality of forecasting is introduced by Sánchez (2008). With a two-step, time-variable combination method, Sánchez achieves for empirical data a forecast quality that yields slightly better results for the given time-period than the best individual forecast.

The goal of the development of the method presented in this article is a forecasting quality that on the average is clearly better than that of the best individual forecast. At the same time, in regard to its practical use, the method should be comprehensible and easy to implement as well as applicable for a longer period of time without manual operator intervention. As a basis the variance–covariance approach turns out to be well suited. Without having to resort to larger adaptations of the standard method, it can be shown using empirical data that by means of a combination of the wind power forecasts, in comparison to the best individual forecast, a clear improvement in the forecasting quality is possible, if the individual forecasts and the estimation period are carefully chosen. In addition, as will be shown in the following, the variance–covariance method can be employed if data gaps in the underlying individual forecasts exist. Moreover, on this basis an improvement in the forecasting quality is possible through an integration of the real-time values. A consideration of the costs on the basis of the forecast volatility in relation to forecast horizon is also realisable through an adaptation. With only a slightly deterioration in the mean squared error, the developed method can succeed in distinctly reducing the volatility.

3.2 Empirical data used

For the development of the method, individual forecasts from several forecast systems, the actual-values of the wind power generation and the installed wind capacity for the period since January 2008 were available for the sum of all wind power plants in Germany in quarter-hourly resolution. Three wind power forecasts of the systems available on the market were chosen so that they covered the broadest spectrum on weather and power forecast models. Each of the three forecast systems (calculated by the companies Energy & Meteo Systems GmbH, EuroWind GmbH and Meteorologica S.A) delivers four to five forecasts a day, depending on the

underlying weather models, for the current day and for the three following calendar days. For the method developed, the most recent forecast in each case is used. For the preparation of the data, the actual and the forecast values are, as a first step, normalised by means of the installed capacity at the respective points of time in the past. The degree of utilisation of the installed capacity at the given point results from a division of the power generation time series by the installed capacity. Afterwards, the values of the degree of utilisation at all times lie in an interval between 0 and 1.¹⁰ This guarantees the comparability of the data at various points of time despite the expansion of capacity (see Sánchez 2008, p. 680).

For the method for combining individual forecasts, data from two successive time periods, $t = 1, \dots, T$ and $t = T + 1, \dots, \hat{T}$ are used. The first time period is employed for the estimation of weighting factors. For the second time period these weights are used for all points for the calculation of a combined forecast. The resulting combined forecast then serves for the comparison of various forecast methods. It can be empirically shown that an estimate using the data of the given previous calendar half-year yields good results, since a calendar half-year represents the shortest time period in which normally all of the possible weather conditions, such as strong and weak wind conditions, are included. All of the represented results in this article are calculated on the basis of the data from the first half of 2011. The corresponding weighting factors are estimated on the basis of the second half of 2010.

The normalised actual-values for a point of time t are referred to as y_t , the accompanying vector of the actual-values $(y_1, \dots, y_T)'$ as $\mathbf{y}$. In each case the most recent individual forecast of a system s ($s = 1, \dots, S$), which is available for a combined forecast $p_{t,h}$ having the forecast horizon $h = 0, 1, \dots, H$, is expressed by $f_{t,s,h}$. On the basis of the given data in each case the horizon h has reference to a quarter-hour. The maximum forecast horizon H amounts to 160 and thus 40 h. This is sufficient for the trading represented in Sect. 2.2. Thus a matrix $\mathbf{F}_h$ of the individual forecasts results in

$$\mathbf{F}_h = \begin{pmatrix} f_{1,1,h} & \cdots & f_{1,S,h} \\ \vdots & \ddots & \vdots \\ f_{T,1,h} & \cdots & f_{T,S,h} \end{pmatrix}. \quad (1)$$

A peculiarity of the forecast method developed here is represented by the fact that in practice temporary outages in individual forecasting systems s repeatedly occur. This results in the absence of some values $f_{t,s,h}$. In order to avoid a bias of the forecast through missing values, it is necessary to introduce binary parameters $b_{t,s,h}$, with

¹⁰ The values 0 and 1 here describe the merely theoretically possible lower and upper limits of the degree of utilisation of the installed capacity of a plant. Since as a rule there is never a complete absence of wind throughout the Federal Republic of Germany, the empirical minimum level lies at just under 0.05. Moreover, the upper limit, at just under 0.85 empirically, clearly lies under the theoretical maximum. The reason for this difference is that there is never a perfect wind situation for all plants at the same time, and on the basis of maintenance measures or defects a certain number of plants will always be at a standstill.

$$b_{t,s,h} = \begin{cases} 1, & \text{if values for } f_{t,s,h} \text{ available} \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

In analogy to the matrices of the individual forecasts a corresponding matrix $\mathbf{B}_h$ results. The forecast error of an individual forecast $e_{t,s,h}$ is thus by means of

$$e_{t,s,h} = \begin{cases} f_{t,s,h} - y_t, & \text{if values for } f_{t,s,h} \text{ available} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

given with the corresponding matrix of the individual forecast errors $\mathbf{E}_h$. The individual forecast errors can be used for the measurement of the forecast quality of the individual forecasts. From the multiplication of the transposed matrix $\mathbf{E}_h'$ with the matrix $\mathbf{E}_h$, a symmetric $S \times S$ matrix $(\mathbf{E}_h' \mathbf{E}_h)$ results. On the basis of the missing data, $(\mathbf{E}_h' \mathbf{E}_h)$ is normalised element wise by the symmetrical matrix $(\mathbf{B}_h' \mathbf{B}_h)$. Thus the symmetric $S \times S$ matrix results in:¹¹

$$\mathbf{K}_h = \begin{pmatrix} \frac{(\mathbf{E}_h' \mathbf{E}_h)_{1,1}}{(\mathbf{B}_h' \mathbf{B}_h)_{1,1}} & \dots & \frac{(\mathbf{E}_h' \mathbf{E}_h)_{1,S}}{(\mathbf{B}_h' \mathbf{B}_h)_{1,S}} \\ \vdots & \ddots & \vdots \\ \frac{(\mathbf{E}_h' \mathbf{E}_h)_{S,1}}{(\mathbf{B}_h' \mathbf{B}_h)_{S,1}} & \dots & \frac{(\mathbf{E}_h' \mathbf{E}_h)_{S,S}}{(\mathbf{B}_h' \mathbf{B}_h)_{S,S}} \end{pmatrix}. \quad (4)$$

On the main diagonal this matrix contains the mean squared error (MSE) of the individual forecasts of a system s . As a measured quantity for the assessment of the quality of wind forecasts, the square root of the MSE, the RMSE, is commonly used (see, for example, Giebel et al. 2011). Often on the basis of the normalisation of the data the designation NRMSE is also used and the value expressed in per cent. Figure 10 shows the NRMSE derived from $\mathbf{K}_h'$ in dependence on h for the validation period.

Figure 10 illustrates that the NRMSE clearly increases in relation to the forecast horizon. Moreover, the relative forecast quality of the systems differs depending on the forecast horizon. The NRMSE of System 3 is in the forecast horizon of 24 quarter hours, for example, clearly worse than that of System 1, but in the forecast horizon of 160 quarter hours the forecast quality of the systems approach each other. System 2 is for every forecast horizon the best system. Thus it will be used in the following discussion as a reference for the measurement of the forecast quality of the forecast combination.

3.3 Simple combination in consideration of data gaps

In the first step of the combination of the wind power forecasts, a method modelled after the classic variance–covariance method of Bates and Granger (1969) will be introduced. On the basis of the differing relative forecast quality of the individual

¹¹ It is $(\mathbf{E}_h' \mathbf{E}_h)_{r,s}$ the element of r -th row and s -th column of the matrix products $(\mathbf{E}_h' \mathbf{E}_h)$ with $r, s = 1, \dots, S$. Moreover $(\mathbf{B}_h' \mathbf{B}_h)_{r,s} \neq 0$ is for all s and r assumed, i.e., at least at one time t , valid values must exist for s and r .

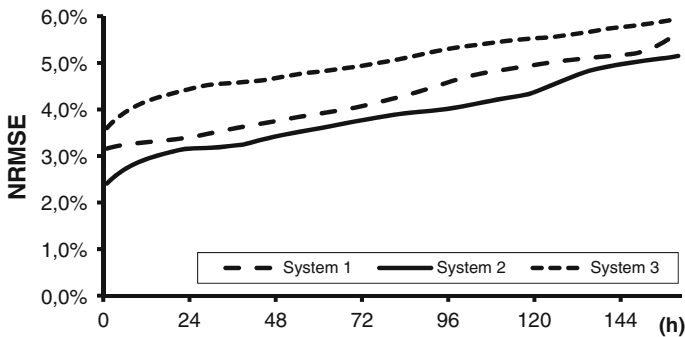


Fig. 10 Forecast quality of the individual forecasts for the first half of 2011

forecasts, the weighting factors are calculated for each forecast horizon h . In the variance–covariance method it is assumed that all the individual forecasts are unbiased with the expected value of the individual forecast errors $E(e_{t,s,h}) = 0$. Likewise the expected value of the error of the individual forecasts F_h combined with a vector w_h of the weighting factors $w_{s,h}$ with $w_h = (w_{1,h}, \dots, w_{S,h})'$ is assumed to be zero. This is generally valid only on the condition that the sum of the weightings is equal to one, that is $w_h \in \mathbb{W} = \{w_h \in \mathbb{R}^S | w_h' I = 1\}$ (see Timmermann 2006, p. 159 und Thiele 1992, p. 20).¹² Under these conditions an optimal weighting vector in which the combined forecast $F_h w_h$ exhibits the smallest squared error compared to the actual-values y can be determined solely on the basis of the covariance matrix of the individual forecast errors.

Since the empirical covariance matrix cannot be directly calculated because of the data gaps for the used data, an estimate of the optimal weighting with the aid of the defined matrix K_h is carried out:

$$\min_{w_h \in \mathbb{W}} (w_h' K_h w_h) \quad (\text{Model 1})$$

Under the assumption that the matrix K_h is invertible, the optimal solution w_h^* from Model 1 can be specified in closed-form (see Dickinson 1975, pp. 205ff.):

$$w_h^* = \frac{K_h^{-1} I}{I' K_h^{-1} I} \quad (5)$$

Model 1 thus serves for the estimate of the optimal weightings w_h^* for all planning horizons ($h = 1, \dots, H$). The estimated weighting factors can then be used for the calculation of the combined forecast for the validation period and of the NRMSE for the respective forecast horizon h .

In Fig. 11 on the basis of Model 1 a clear improvement of the forecast quality is recognisable. Depending on the forecast horizon, the NRMSE improves in comparison to the best individual forecast between 8 and 15 %. Mainly for the

¹² When the total historical time frame is examined, the individual forecasts are almost completely unbiased. The very small bias that is to be noticed in the estimation time frame is not systematic and is not repeated in the validation time frame. Therefore, the condition $w_h' I = 1$ is reasonable in making the estimate.

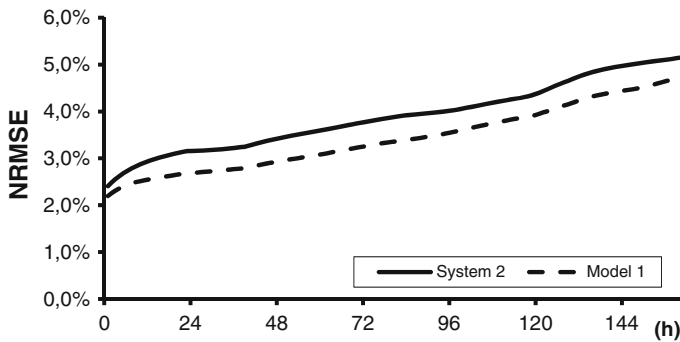


Fig. 11 Forecast quality of the forecast combination in accordance with Model 1

medium-term forecast horizons between 30 and 100 quarter-hours, the forecast quality increases at an above-average rate. For the short-term forecast horizons that are especially significant for the minimising of the use of balancing energy, however, only a below-average improvement results from the presented approach. In the next step the method will thus be enhanced especially for the very short-term horizon.

3.4 Forecasts using real-time values

Starting from any point of time $\bar{t}$, the most recent forecast of a system s delivers the forecast values $\hat{f}_{\bar{t}+h,s,h}$ $h = 0, 1, \dots, H$. Hence the appropriate time series of the individual forecast errors $e_{\bar{t}+h,s,h}$ for each system s can be directly determined. In these time series it is immediately apparent that the individual forecast errors $e_{\bar{t}+h,s,h}$ of a system are not independent of each other, since the time series, for example, exhibit up and down movements over several points of time. An analysis of the relation between the individual forecast errors depending on a lag is possible using an autocorrelation function. The calculated empirical autocorrelation functions from the time series of the individual forecast errors of the validation time period are represented in Fig. 12 for the three systems employed in dependence on a lag of a quarter-hours. It can be seen that the autocorrelation between the individual forecast errors analysed becomes weaker with a growing lag a . The small increase of autocorrelation in the area of 96 quarter-hours is linked with the daily course of some of the weather phenomena, such as the sea-breeze. Moreover, Fig. 12 makes it obvious that in respect to the autocorrelation, there are differences between the systems.

The courses of the autocorrelation functions, which up to an a of 60 quarter hours almost decline geometrically, indicate an autoregressive process whose parameters differ in relation to s . Following the approach of Box and Jenkins (see Box et al. 2008, pp. 195ff. or Rinne and Specht 2002, pp. 375ff.) the empirical partial autocorrelation functions of the individual forecast errors represented in Fig. 13 will be studied for further model identification. These functions feature in each case a

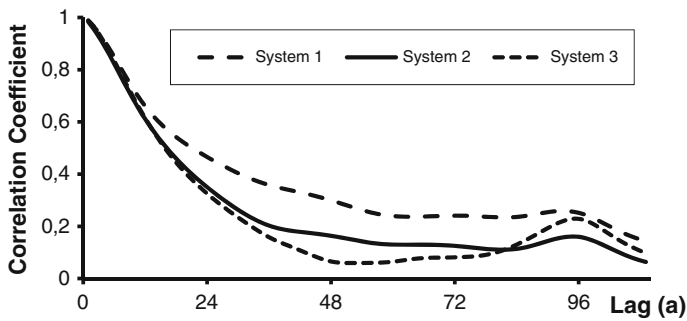


Fig. 12 Autocorrelation functions of the individual forecast errors

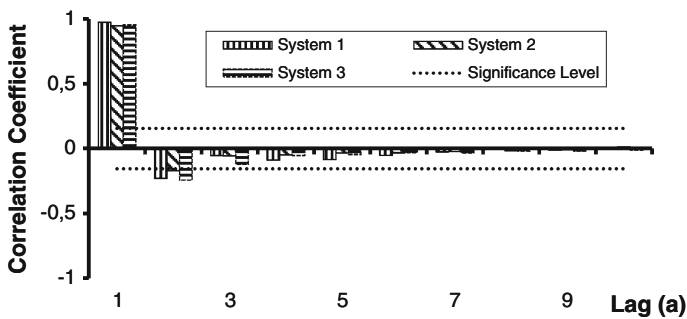


Fig. 13 Partial autocorrelation functions of the individual forecast errors

strongly significant value with a lag of 1 as well as an additional weakly significant value with a lag of 2. For this reason an interpretation of the individual forecast error time series as AR(2) or AR(1) process is possible. For System 2, for example, an AR(2) process with the weights φ_1 of 1.369 and φ_2 -0.378 leads to the result, that for the remaining residuals, neither the autocorrelation function nor the partial autocorrelation function shows significant values. An explanation of the error time series by means of an AR(1) process with a weight φ_1 of 0.958 yields good results through the example of System 2 as well. Up to a lag of 8 a weak autocorrelation, however, remains in the residuals. With the other systems the situation is comparable. Only the weights differ in comparison to the example presented.

For the further development of the forecast method an AR(1) and an AR(2) approach were taken into consideration. However, for the validation time period the forecast quality of the final model in the two approaches does not differ. In accordance with the axiom of parsimony (Box et al. 2008, p. 16), therefore, an AR(1) approach was referred to for the further development of the forecast method.

On the basis of the interpretation of the individual forecast-error time series as autoregressive process and the availability of the real-time values, an improvement

in the forecast quality of the individual forecasts is possible. This can be achieved by an autoregressive forecast of the error time series which is used for the correction of the individual forecasts. A forecast of the error time series $e_{\bar{t}+h,s,h}$ based on $h = 0$ with $e_{\bar{t}+h,s,h} = e_{\bar{t}+0,s,0}(\varphi_s)^h$ yields yet unsatisfactory results for $h > 60$ because of the following increase in the autocorrelation (see for the forecast of autoregressive processes, for example, Box et al. 2008, pp. 164ff). Better results occur by means of an individual estimate of weights $\varphi_{s,h}$ for each horizon h .

The goal of the developed approach is not primarily an improvement of the quality of the individual forecasts, but rather the best possible combined forecast. For the empirical data used, a simultaneous estimate of the weights $\varphi_{s,h}$ at once with a combination of the forecasts consequently yields better results than does one individual estimate per system with a subsequent combination. For a simultaneous estimate the variants $f_{t,s,h}^{abs}$ and $f_{t,s,h}^{rel}$ of the individual forecasts are generated:

$$f_{\bar{t}+h,s,h}^{abs} = f_{\bar{t}+h,s,h} - e_{\bar{t}+0,s,0} \quad \forall \bar{t}, s, h \quad (6)$$

$$f_{\bar{t}+h,s,h}^{rel} = f_{\bar{t}+h,s,h} \cdot \left(\frac{y_{\bar{t}+0}}{f_{\bar{t}+0,s,0}} \right) \quad \forall \bar{t}, s, h \quad (7)$$

The forecast variant $f_{t,s,h}^{abs}$ corresponds to the initial forecast $f_{t,s,h}$ that was adjusted by the absolute forecast error at the time $\bar{t}$ with the aid of the real-time value, as represented in Eq. (6). Likewise for the variant $f_{t,s,h}^{rel}$ an adjustment on the basis of observed relative forecast errors—as in Eq. (7)—takes place. Depending on the forecast system one of the variants yields better results.

The calculated forecast-variants $f_{t,s,h}^{abs}$ and $f_{t,s,h}^{rel}$ are now combined together with the original forecasts. Thus for the forecast combination there are, with the original forecasts and the respective two variants, $\tilde{S} = S * 3$ forecasts available. The corresponding symbols of the weighting vectors and matrices, which also contain the forecast variants, are marked with a tilde ($\sim$) in the following. Through the consideration of the forecast variants in the combination, the weights $\varphi_{s,h}$ follow, implicitly through the weights $\tilde{w}_{s,h}$. High weights of the forecast variants in comparison to the original forecasts thus corresponds to a high $\varphi_{s,h}$, low weights of the forecast variants to a low $\varphi_{s,h}$.

With the introduction of the forecast variants, the problem of multicollinearity in the estimation of the weighting factors becomes more important. Since the individual forecasts and especially the forecast variants have a very high correlation among each other, the weighting factors are often unstable with regard to small changes in the estimation data. Moreover, in the combination of forecasts—in contrast to Model 1—despite the positive correlation between the individual forecasts and the actual-values, negative weighting factors can occur, as in the case of a large number of forecasts often happens (see Thiele 1992, pp. 71ff.). In such a case the introduction of linear constraints can improve the stability of the estimated weighting factors (see Clemen 1986, pp. 32ff.) and can moreover prevent unintended results, such as negative weighting factors. Therefore in the following,

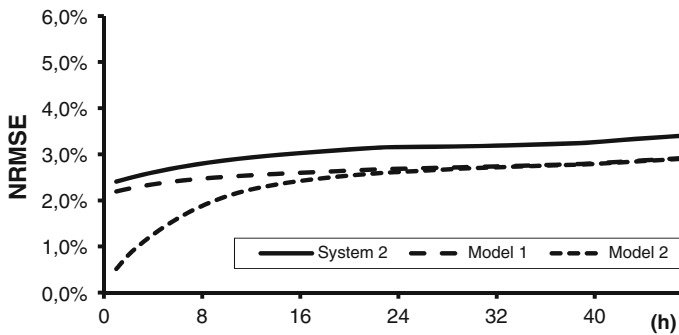


Fig. 14 Forecast quality of the forecast combination Model 2

for the combination of the forecasts, nonnegativity constraints are added, i.e. $\tilde{\mathbb{W}}_+ = \{\tilde{\mathbf{w}}_h \in \mathbb{R}_+^{\tilde{S}} | \tilde{\mathbf{w}}_h' \mathbf{1} = 1\}$ and the following optimisation problem arises:

$$\min_{\tilde{\mathbf{w}} \in \tilde{\mathbb{W}}_+} (\tilde{\mathbf{w}}_h' \tilde{\mathbf{K}}_h \tilde{\mathbf{w}}_h) \quad (\text{Model 2})$$

A closed-form solution of the optimisation problem is after the introduction of the nonnegativity constraints no longer readily possible. Model 2 is, however, solvable as a quadratic optimisation problem (QP).¹³ For the chosen validation period the combined forecast $\tilde{\mathbf{p}}_h^*$ results through the optimal weighting factors $\tilde{\mathbf{w}}_h^*$.

In Fig. 14 it can be seen that for Model 2 a major improvement in the NRMSE in the short-term time horizon up to about thirty quarter hours occurs. For larger forecast horizons no significant improvement can be achieved.

3.5 Combination in consideration of the costs of the forecasting fluctuations

The determination of the weighting factors in Models 1 and 2 is carried out in conformity with the variance–covariance method only on the basis of the mean squared error. For an optimally combined forecast with regard to the costs of trading RES-E, however, the intensity of the fluctuations of the combined forecast for the same point of time t with different forecast horizons h must be considered. If the fluctuations are high, they will cause a large trading volume on the intraday market with the effects on the costs characterised above (see Fig. 8). From an economic point of view it is not sufficient to determine the weighting of the forecasts only by means of the mean squared error. In this section, therefore, an extension of the previous approach through the additional consideration of the fluctuations of the forecasts and the resulting costs will be made.

¹³ For example, the Frank–Wolfe algorithm can be used (see, Domschke and Drexl 2011, pp. 195ff.). However, the use assumes a convex objective function, that is, that the matrix $\tilde{\mathbf{K}}_h$ must be positive semidefinite. From the derivation of $\tilde{\mathbf{K}}_h$ it is clear that this is not always guaranteed when data gaps occur. For the empirical data used in the process this, however, was always the case. Otherwise, various methods may serve to generate a ‘most similar’ positive semidefinite matrix (see, for example, Higham 2002).

If a constant change in the forecasts with decreasing forecast horizon is desirable, the volatility is a good indicator for a quick comparison of different forecast models with regard to the intensity of the fluctuation. The volatility of the combined forecast $p_{t,h}$ is specified as the standard deviation of the forecast changes $p_{t,h} - p_{t,h-1}$. The average volatility of the original individual forecasts of the three systems amounts to between 0.38 and 0.41 % of the installed capacity. Through the combination with Model 1, the volatility drops to 0.25 % in addition to the represented improvement of the NRMSEs. Despite the further improvement of the NRMSEs through Model 2, however, the volatility here rises again to 0.31 %. This is caused by, among other matters, the actual-values which are volatile in themselves and enter the model as additional information through the forecast variants. Here, for example, a volatility of the combined forecast can result, even if the individual forecasts with respect to h are constant. Moreover, the weighting factors differ more than in Model 1 through the increased amount of individual forecasts in dependence on the forecast horizon.

The modelling of the costs caused by the fluctuations of the forecast for the determination of the weighting factors will be presented in the following. Regarding the correlation shown in Sect. 2.3 between the amount of a forecast difference and the costs of intraday trading, a quadratic loss function is assumed for the modelling. It is presumed that the combined forecast $p_{t,h}$ corresponds to the state of the trading at a point in time $t - h$. Later in time at the point $t - h + n$, forecasts $p_{t,h-n}$ are available. Hereby n ($n = 1, \dots, h - 1$) indicates the temporal difference between two forecasts. The costs of the intraday trading then result basically from the squared difference of the combined forecasts $(p_{t,h} - p_{t,h-n})^2$.

However, it is to be noted that a difference in forecast involves a transaction on the intraday market only if the trading volume exceeds a minimum size m ($m > 0$).¹⁴ To be able to estimate the costs of the intraday trading, the probabilities P_n^{id} are of interest whether a forecast difference causes a transaction. A corresponding transaction occurs as soon as the difference between the forecasts $p_{t,h}$ and $p_{t,h-n}$ is for the first time larger than m , i.e., when $|p_{t,h} - p_{t,h-n}| > m$ is valid. The prerequisite is thus that in the time frame between $t - h$ and $t - h + n - 1$ no transaction was carried out. On the basis of the on-going actualisations of the individual forecasts and the changes in the actual-values, the probability declines with increasing n that no transaction between $t - h$ and $t - h + n - 1$ will be carried out. By means of the calculated combined forecast in Model 2, this probability can be approximated for all n with the aid of an empirical frequency $\bar{P}_n = A(n) = b^n$. The empirical data shows that on the basis of forecasts for all forecast horizons h , an approximation of a constant b is possible. In Fig. 15 the empirical frequencies of $\bar{P}_n$ für $m = 0.5$ % of the installed capacity, $b = 0.925$ and for $h = 160$, are exemplified.

¹⁴ The introduction of a minimum trading volume has various operational reasons. For one thing, on the European Power Exchange EPEX a minimum volume of 0.1 MWh per transaction exists. Moreover, operational issues, such as the temporal need for a transaction or transaction costs, play a role that in addition demand a higher minimum volume. A minimum trading volume of 0.5 % of the installed capacity means for a company that markets 15 % of the nationwide wind energy an absolute value of approx. 20 MW and thus an average transaction value of approx. 1,200 €.

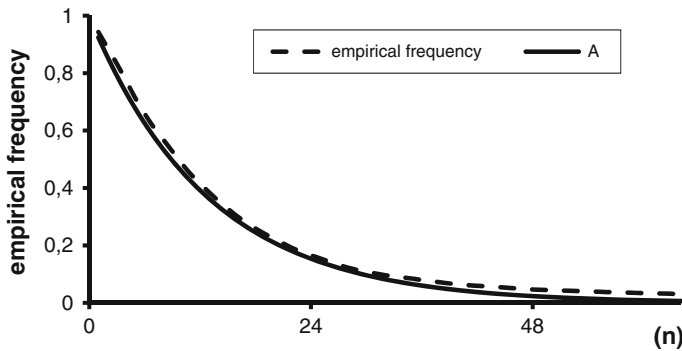


Fig. 15 Probability and empirical frequency of no transactions

The probability P_n^{id} for a transaction of the kind $p_{t,h} - p_{t,h-n}$ now results as $P_n^{id} = \bar{P}_{n-1} - \bar{P}_n$. The farther both forecast points are separated from each other, the smaller the probability. The costs of a forecast difference can thus be estimated as follows: $P_n^{id} \cdot (p_{t,h} - p_{t,h-n})^2$. For the entire estimation time period the average costs of the intraday trading c^{id} thus result in the sum of $P_n^{id} \cdot (p_{t,h} - p_{t,h-n})^2$ for all t and h and all $n < h$ divided by the number of all hours T . Analogous to the formulation of Model 1 and Model 2, c^{id} can under the condition $\tilde{\mathbf{w}}'_h \mathbf{I} = 1$ be estimated by means of the individual forecast errors $\tilde{\mathbf{E}}_h$ (see appendix):

$$c^{id} \approx \sum_{h=1}^H \sum_{n=1}^{h-1} P_n^{id} \cdot (\tilde{\mathbf{w}}'_h \tilde{\mathbf{K}}_h \tilde{\mathbf{w}} + \tilde{\mathbf{w}}'_{h-n} \tilde{\mathbf{K}}_{h-n} \tilde{\mathbf{w}}_{h-n} - 2\tilde{\mathbf{w}}'_h \tilde{\mathbf{L}}_{h,h-n} \tilde{\mathbf{w}}_{h-n}) \quad (8)$$

As demonstrated in Sect. 2.3, besides the costs for intraday transactions in the trading of RES-E, costs for balancing energy also occur. Likewise, in the case of balancing energy, a quadratic correlation between the amount of a transaction and their costs can be verified. Since the costs for the balancing energy are decidedly higher than those of intraday trading, they are illustrated in the following by a larger gradient in the cost function with the factor α . The probability that forecast errors can be offset by balancing energy can be derived from $\bar{P}_n$. Based on a forecast $p_{t,h}$, P_h^{ae} poses for $n = h - 1$ the probability that the use of balancing energy of the nature $p_{t,h} - y_t$ takes place. Thus the costs of balancing energy based on a forecast $p_{t,h}$ can be estimated as follows: $\alpha P_h^{ae} (p_{t,h} - y_t)^2$.

For the total estimation time period the average costs of the use of the balancing energy c^{ae} are calculated as follows:

$$c^{ae} \approx \sum_{h=1}^H \alpha P_h^{ae} (\tilde{\mathbf{w}}'_h \tilde{\mathbf{K}}_h \tilde{\mathbf{w}}_h) \quad (9)$$

The determination of the weighting factors with minimal costs

$$\min_{\tilde{\mathbf{w}} \in \tilde{\mathbb{W}}_+} (c^{ae} + c^{id}) \quad (\text{Model 3})$$

can thus by aid of the Eqs. 8 and 9 be carried out. For the validation time period the combined forecast results from the optimum weighting factors in Model 3, which

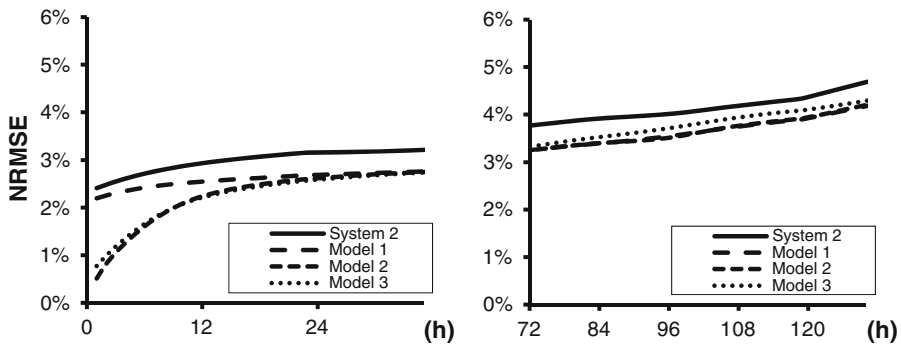


Fig. 16 Forecast qualities in comparison

were calculated by an α with the value of 3. The resulting forecasting quality in comparison to the other models is represented in Fig. 16.

Compared with Model 2 there is in Model 3 on the whole only a slight decline in the NRMSE. At the strongest it comes out in the range of a forecast horizon of less than 12 quarter hours (left graph, Fig. 16) and in the area of the medium forecast horizon between 72 and 136 quarter hours (right graph, Fig. 16). The volatility, in contrast, clearly declines to 0.24 % and thus lies even below the volatility of Model 1. Looking at the weighting factors, two main reasons for the reduction of the volatility can be recognised. In Fig. 17 these are represented in comparison for Model 2 and Model 3. Here the above two graphs show the weighting factors of the original individual forecasts $f_{i,s,h}$, the lower two graphs show the weighting factors of the forecast variants $f_{i,s,h}^{abs}$ corrected by the absolute forecast errors. The change in the weighting factors depending on the forecast horizon h can be recognised in all four graphs. Thus in Model 3 (the graph above and to the right), for example, up to a horizon of just under 70 quarter hours System 2 is the most heavily weighted, for greater horizons it is System 3. The fluctuation of the weighting factors can be explained, for example, by a change in the relative forecast quality of the systems in relation to each other. In the lower two graphs the weighting of the relative forecasting errors declining in accordance with h can be easily noticed. This corresponds to the declining autocorrelation with regard to h . A comparison of the forecast weighting of Model 2 and Model 3 illustrates that the fluctuations of the weighting factors depending on h are considerably smaller in Model 3 than in Model 2. This has the largest influence on the reduction of the volatility. Moreover, it is conspicuous that the original forecast in Model 3 receives in total a somewhat higher weighting than in Model 2. The result is that the fluctuations of the real-time-values have a smaller influence on the combined forecast of Model 3 than in Model 2.

In addition to the comparison of the volatility of the different models, a comparison of the models by means of the loss function values is also possible, as it is used in Model 3. Since the absolute value of the loss function has only a modest significance, a comparison relative to the value of the loss function for System 2 follows. On the basis of the best individual forecast, System 2, the loss function

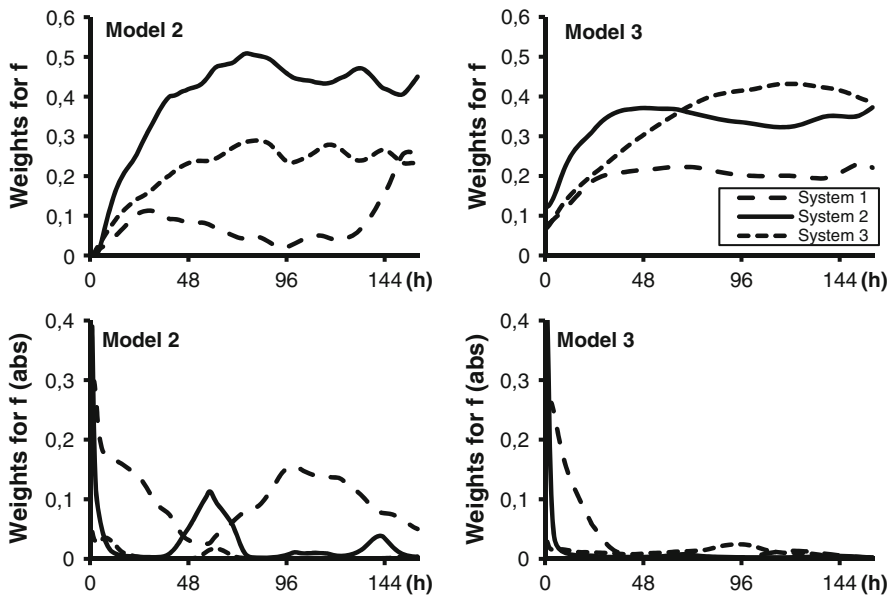


Fig. 17 Weighting factors of Model 2 and 3 in comparison

value receives in the following the value of 100 %. In Model 1 the value declines to 62.7 %. In Model 2 the value increases despite clearly better NRMSE slightly to 65.2 %. Model 3, on the other hand, lowers the loss function value to 52.4 %. Therefore, through the combination of the forecasts the costs in comparison to the best individual forecast are in total reduced by about almost 50 %.

4 Conclusions

The growth of electricity generation through renewable energies has led to a change in the German electricity market and respective to the demands on the market players. The trading of RES-E differs from trading electricity of conventional generation. In all possible variants of the trading of fluctuating renewable energy in accordance with the EEG, the forecast employed plays a significant role for the level of trading-costs. In this article the conditions for the trading of RES-E in Germany were described. In other countries these conditions can differ on the basis of the support schemes for renewable energy. Nevertheless, there is from an energy-economics point of view a similar influence of the forecasting quality on the marketing costs if comparable market mechanisms, such as an intraday market with continuous trading, are to be found. A transfer of the results to other countries is possible if existing differences are taken account of.

For companies carrying out the trading of RES-E, the method presented here for the combination of forecasts provides an excellent instrument of improving the quality of forecasting and thus reducing the trading costs. Both forecast error as

ascertained by the NRMSEs and also the volatility of the forecasts with respect to the forecast horizon can be clearly improved by applying this method. In comparison to the model presented here, a further improvement of the quality of forecasting is possible if additional weighting factors that are determined by means of the data of a very short time period prior to the forecasts are integrated in the model. A system that contains these enhancements but otherwise basically depends on the method described here was developed by the authors for TransnetBW GmbH. There the system is currently in operation for the marketing of a good 14 % of the nationwide wind electricity generation. In comparison to the systems used by the other three German TSOs, very good results have been achieved.¹⁵

The current transformation in the generation of electricity from fossil or nuclear energy sources to renewable energies will continue to entail enormous challenges. Through the increasing market integration, the trading of renewable energies will greatly gain in importance, not least from a business perspective. The business-related research in this field, however, has hardly advanced beyond its infancy and will find itself faced with a host of fascinating questions.

Appendix

Under the introduction of an index r that is equivalent to index s and the condition $\tilde{w}'_h \mathbf{1} = 1$, formula 8 can be derived as follows. In this case it is first assumed that at no point in time are forecasting data missing

$$\begin{aligned}
 c^{id} &= \frac{1}{T} \sum_{h=1}^H \sum_{n=1}^{h-1} P_n^{id} \sum_{t=1}^T (p_{t,h} - p_{t,h-n})^2 \\
 &= \frac{1}{T} \sum_{n=1}^{h-1} P_n^{id} \sum_{t=1}^T \left(y_t + \sum_{s=1}^{\tilde{S}} w_{s,h} e_{t,s,h} - \left[y_t + \sum_{s=1}^{\tilde{S}} w_{s,h-n} e_{t,s,h-n} \right] \right)^2 \\
 &= \frac{1}{T} \sum_{h=1}^H \sum_{n=1}^{h-1} P_n^{id} \sum_{t=1}^T \left(\sum_{s=1}^{\tilde{S}} w_{s,h} e_{t,s,h} - \sum_{s=1}^{\tilde{S}} w_{s,h-n} e_{t,s,h-n} \right)^2 \\
 &= \frac{1}{T} \sum_{h=1}^H \sum_{n=1}^{h-1} P_n^{id} \sum_{s=1}^{\tilde{S}} \sum_{r=1}^{\tilde{S}} \left(w_{s,h} w_{r,h} \sum_{t=1}^T e_{t,s,h} e_{t,r,h} + w_{s,h-n} w_{r,h-n} \sum_{t=1}^T e_{t,s,h-n} e_{t,r,h-n} \right. \\
 &\quad \left. - 2 w_{s,h} w_{r,h-n} \sum_{t=1}^T e_{t,s,h} e_{t,r,h-n} \right)
 \end{aligned}$$

¹⁵ A comparison of the systems is relatively straightforward, since all four TSOs merchandise a certain percentage of the total German wind energy generation, and a large part of the data regarding the merchandising is published in the Internet.

In the case of missing forecast data an estimate of the costs with the aid of the matrix $\tilde{L}_{h,h-n}$ is possible:

$$c^{id} \approx \sum_{h=1}^H \sum_{n=1}^{h-1} P_n^{id} \cdot (\tilde{\mathbf{w}}'_h \tilde{\mathbf{K}}_h \tilde{\mathbf{w}} + \tilde{\mathbf{w}}'_{h-n} \tilde{\mathbf{K}}_{h-n} \tilde{\mathbf{w}}_{h-n} - 2\tilde{\mathbf{w}}'_h \tilde{L}_{h,h-n} \tilde{\mathbf{w}}_{h-n})$$

As a result it is valid for all h and all $n < h$:

$$\tilde{L}_{h,h-n} = \begin{pmatrix} \frac{(\tilde{E}'_h \tilde{E}_{h-n})_{1,1}}{(\tilde{B}'_h \tilde{B}_{h-n})_{1,1}} & \dots & \frac{(\tilde{E}'_h \tilde{E}_{h-n})_{1,\tilde{S}}}{(\tilde{B}'_h \tilde{B}_{h-n})_{1,\tilde{S}}} \\ \vdots & \ddots & \vdots \\ \frac{(\tilde{E}'_h \tilde{E}_{h-n})_{\tilde{S},1}}{(\tilde{B}'_h \tilde{B}_{h-n})_{\tilde{S},1}} & \dots & \frac{(\tilde{E}'_h \tilde{E}_{h-n})_{\tilde{S},\tilde{S}}}{(\tilde{B}'_h \tilde{B}_{h-n})_{\tilde{S},\tilde{S}}} \end{pmatrix}.$$

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Legal texts

- AusglMechAV Verordnung zur Ausführung der Verordnung zur Weiterentwicklung des bundesweiten Ausgleichsmechanismus. <http://bundesrecht.juris.de/bundesrecht/ausglmechav/gesamt.pdf>.
- AusglMechV Verordnung zur Weiterentwicklung des bundesweiten Ausgleichsmechanismus. <http://bundesrecht.juris.de/bundesrecht/ausglmechv/gesamt.pdf>.
- EEG Gesetz für den Vorrang Erneuerbarer Energien. http://www.erneuerbare-energien.de/files/pdfs/allgemein/application/pdf/eeg_2012_bf.pdf.
- Einspeisegesetz Gesetz über die Einspeisung von Strom aus erneuerbaren Energien in das öffentliche Netz. <http://archiv.jura.uni-saarland.de/BGBI/TEIL1/1990/19902633.1.HTML>.